Iteration (Isimbi Nelly)

★ Predictions:

$\hat{y}_1 = (-1.3316)(1) + (1.1808)(3) + 0.9318 = 3.1426$

$\hat{y}_2 = (-1.3316)(4) + (1.1808)(10) + 0.9318 = 7.4124$

Predictions: $\hat{y} = [3.1426, 7.4124)$

★ Errors:

$e_1 = 3.1426 - 5 = -1.8574$

$e_2 = 7.4124 - 6 = 1.4124$

Errors: $e = [-1.8574, 1.4124]$

★ Gradients:

For m_1: $(-1.8574)(1) + (1.4124)(4) = 3.7922$

$$\frac{\partial J}{\partial m_1} = 3.7922$$

For m_2= $(-1.8574)(3) + (1.4124)(10) = 8.5528$

$$\frac{\partial J}{\partial m_2} = 8.5528.$$

$-1.8574 + 1.4124 = -0.4450$

$$\frac{\partial J}{\partial b} = -0.4450$$

★ Update Parameters: $m_1 = -1.3316 - 0.01(3.7922) = -1.3695$

$m_2 = 1.1808 - 0.01\ (8.5528) = 1.0953$

$b = 0.9318 - 0.01\ (-0.4450) = 0.9363$

After Iteration $m = [-1.3695,\ 1.0953]$

$b = 0.9363.$

$$e = \hat{y} - y = \begin{bmatrix}1\\11\end{bmatrix}$$

$$= \begin{bmatrix}6\\17\end{bmatrix} - \begin{bmatrix}5\\6\end{bmatrix} = \begin{bmatrix}1\\11\end{bmatrix}$$

Gradient for m $\frac{\partial J}{\partial m} = X^T e$

$$X^T = \begin{bmatrix}1 & 4\\3 & 10\end{bmatrix}$$

$$= \begin{bmatrix}1 & 4\\3 & 10\end{bmatrix}\begin{bmatrix}1\\11\end{bmatrix} = \begin{bmatrix}45\\113\end{bmatrix}$$

Gradient for b $\frac{\partial J}{\partial b} = \begin{bmatrix}1\\11\end{bmatrix} = 12$

update m $= \begin{bmatrix}-1\\2\end{bmatrix} - \begin{bmatrix}0.45\\1.13\end{bmatrix} = \begin{bmatrix}-1.45\\0.87\end{bmatrix}$

update b, $b_1 = b - 0.01 \cdot \cancel{[\;]}\ 12$

$$= \begin{bmatrix}1\\1\end{bmatrix} - \begin{bmatrix}0.01\\0.11\end{bmatrix} = \begin{bmatrix}0.88\\0.88\end{bmatrix}$$

After iteration, $= m = \begin{bmatrix}-1.45\\0.87\end{bmatrix}$ $b = \cancel{\begin{bmatrix}0.99\\0.89\end{bmatrix}}$

$b_2 \begin{bmatrix}0.88\\ \cancel{0.99}\end{bmatrix}$

$b = 0.99$

4. Update Parameters

$$m_1 = -1.45 - 0.01(-11.84) = -1.3316$$

$$m_2 = 0.87 - 0.01(-31.08) = 1.1808$$

$$b = 0.88 - 0.01(-5.18) = 0.9318$$

After Iteration 2

$$m = [-1.3316, 1.1808]$$

Iteration 2 (Sarah)

1. Predictions

$$\vec{y_1} = (-1.45)(1) + (0.87)(3) + 0.88 = 2.04$$

$$\vec{y_2} = (-1.45)(4) + (0.87)(10) + 0.88 = 3.78$$

Predictions

$$\vec{y} = [2.04; 3.78]$$

2. Errors

$$e_1 = 2.04 - 5 = -2.96$$
$$e_2 = 3.78 - 6 = -2.22$$

Errors

$$e = [-2.96, -2.22]$$

3. Gradient

for m_1: $(-2.96)(1) + (-2.22)(4) = -11.84$

$$\frac{\partial J}{\partial m_1} = -11.84$$

for m_2: $(-2.96)(3) + (-2.22)(10) = -31.08$

$$\frac{\partial J}{\partial m_2} = -31.08$$

for b:

$$-2.96 + (-2.22) = -5.18$$

$$\frac{\partial J}{\partial b} = -5.18$$

4. Update parameters

$m_1 = -$

Chain rule: $J \to \hat{y} \to m$

$$\frac{\partial J}{\partial m} = \frac{\partial J}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial m}$$

$J = \frac{1}{2}(\hat{y} - y)^2$ } Differentiate with respect to $\hat{y}$

$\frac{\partial J}{\partial \hat{y}} = \hat{y} - y$

Differentiate prediction: $\hat{y} = xm + b$

$\frac{\partial \hat{y}}{\partial m} = x$

Combine chain rule $\frac{\partial J}{\partial m} = x^T(\hat{y} - y)$

Derivative with respect to bias

Chain rule $\frac{\partial J}{\partial b} = \frac{\partial J}{\partial y} \times \frac{\partial \hat{y}}{\partial b}$ since $= \hat{y} = xm + b$

$\boxed{\frac{\partial J}{\partial b} = \hat{y} - y}$ derivative of $\hat{y}$ with respect to b

Final Gradient $= \left\{ \begin{array}{l} \frac{\partial J}{\partial m} = x^T(\hat{y} - y) \\ \frac{\partial J}{\partial b} = \end{array} \right.$